

Figure 10.4: Electromagnetic Radiation in a Cavity

There is only one important number associated with a harmonic oscillator —namely, its frequency. You probably already know how to calculate the frequency of a wave of length λ:

$$\omega = \frac{2\pi c}{\lambda}.$$

In classical physics, of course, the frequency is just the frequency. But in quantum mechanics, the frequency determines the quantum of energy of the oscillator. In other words, the energy contained in waves of length λ has to be

$$(n + 1/2)\hbar\omega.$$

The term (1/2)*ħω* is not important for our purposes. It's called the zero-point energy, and we can ignore it. If we do, the energy of waves of length λ becomes

$$\frac{2\pi\hbar c}{\lambda}n,$$

where *n* can be any integer from zero on up. In other words, the energy of an electromagnetic wave is quantized in indivisible units of

$$\frac{2\pi\hbar c}{\lambda}.$$